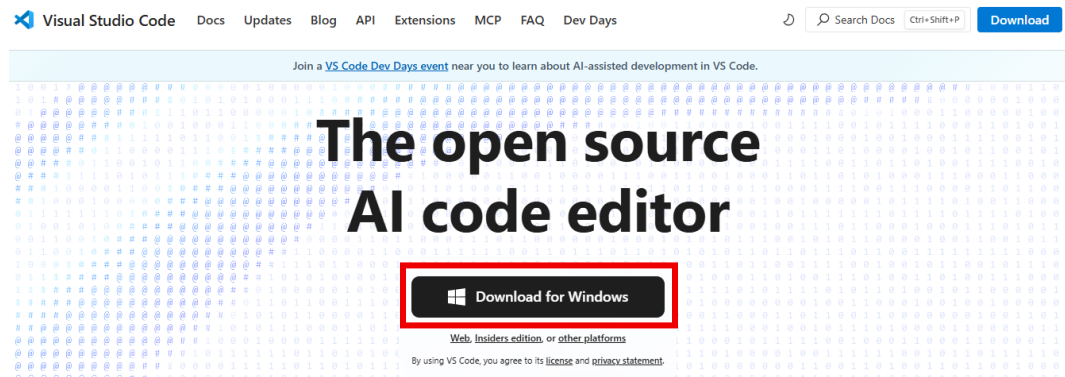


AI Foundry & AI Services Hands on Lab

실습 환경 생성

1. VSCode 설치

- [Visual Studio Code - Code Editing. Redefined](#) 접속
- Download for Windows 클릭하여 다운로드



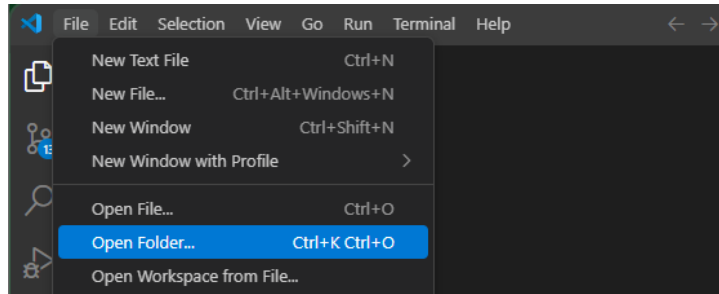
2. Python 설치

- [Welcome to Python.org](#)
- Python 버전 선택 후 (3.13.7 권장) OS 에 맞는 installer 다운로드하여 설치

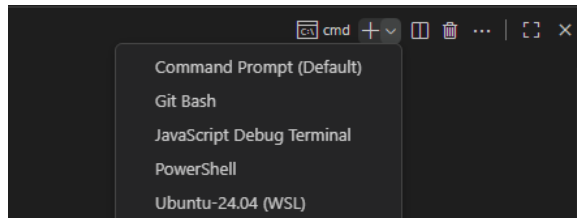
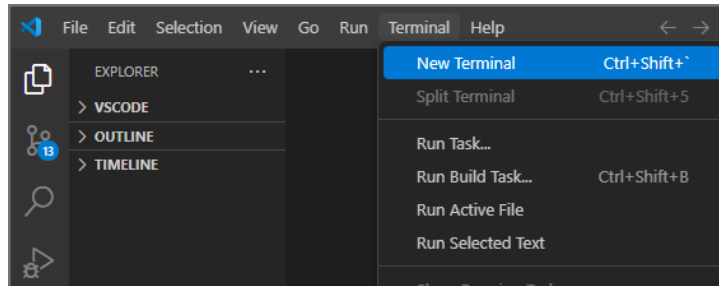
Files							
Version	Operating System	Description	MD5 Sum	File Size	GPG	Sigstore	SBOM
Gzipped source tarball	Source release		138c2e19c835ead10499571e0d4cf189	28.0 MB	SIG	sigstore	SPDX
XZ compressed source tarball	Source release		256cdb3bbf45cdce7499e52ba6c36ea3	21.7 MB	SIG	sigstore	SPDX
macOS 64-bit universal2 installer	macOS	for macOS 10.13 and later	ac0421b04eef155f4daab0b023cf3956	67.8 MB	SIG	sigstore	
Windows installer (64-bit)	Windows	Recommended	1da92e43c79f3d1539dd23a3c14bf3f0	27.5 MB	SIG	sigstore	SPDX
Windows installer (32-bit)	Windows		f501c1b321c82412ed330ec5604cac39	26.2 MB	SIG	sigstore	SPDX
Windows installer (ARM64)	Windows	Experimental	66c0ba98b20b7d4ea0904223d484d369	26.8 MB	SIG	sigstore	SPDX
Windows embeddable package (64-bit)	Windows		77f294ec267596827a2ab06e8fa3f18c	10.4 MB	SIG	sigstore	SPDX
Windows embeddable package (32-bit)	Windows		53eae0de1231fdf133ea703e3e43b30	9.2 MB	SIG	sigstore	SPDX
Windows embeddable package (ARM64)	Windows		6a3fcfc7a10a3822459b1c214f769df1	9.7 MB	SIG	sigstore	SPDX
Windows release manifest	Windows	Install with 'py install 3.13'	6a5588452e73f959c9fa1ba009f42f75	14.6 KB		sigstore	

3. Python venv 설치

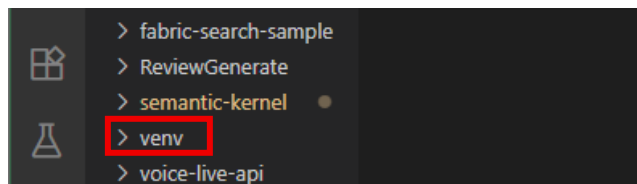
- 프로젝트 폴더 설정 (File > Open Folder)
※ 기존 폴더를 지정해도 되지만, 새로운 폴더 생성 후 사용 권장



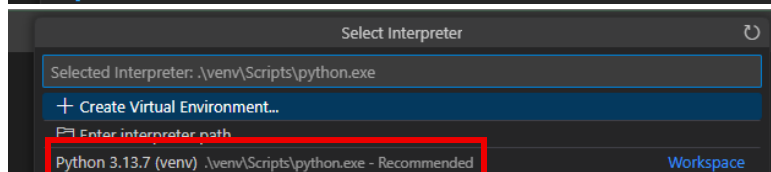
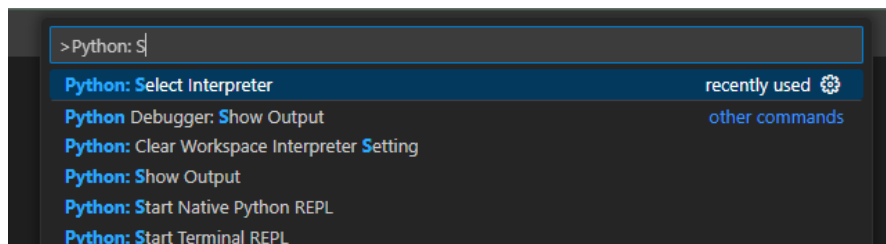
- New Terminal> cmd (Command Prompt) 선택



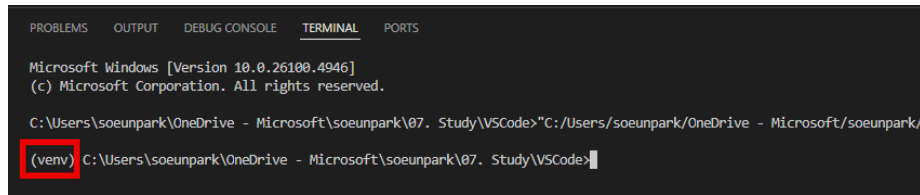
- Terminal 에서 아래 명령어 실행
 - `python -m venv venv`
- venv 생성 확인



- [Ctrl + Shift + P] 키를 눌러 Python: Select Interpreter 검색하여 생성한 venv 선택



- Terminal 재실행 > venv 활성화 확인



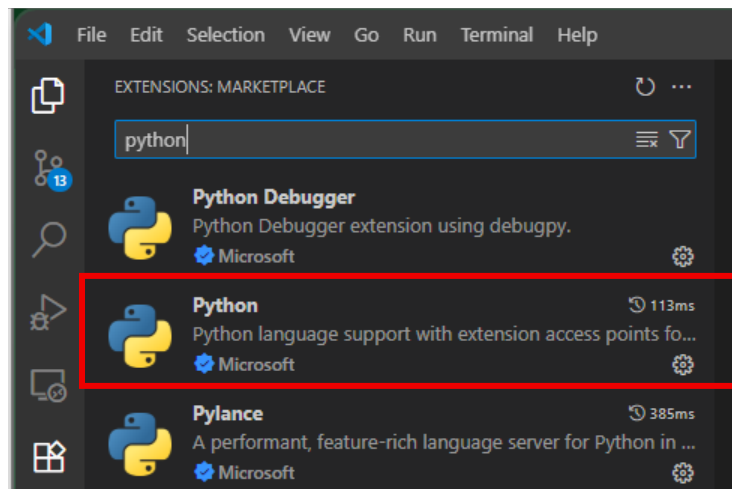
A screenshot of the VS Code terminal interface. The terminal shows the Windows command prompt environment. The prompt is `(venv) C:\Users\soeunpark\OneDrive - Microsoft\soeunpark\07. Study\VSCode>`. The `(venv)` prefix is highlighted with a red box, indicating that the virtual environment is successfully activated.

4. VSCode extension 설치

- 왼쪽 메뉴에서 아래 아이콘 클릭



- Python 선택 후 install

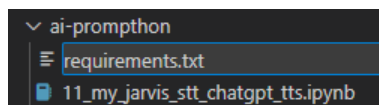


5. 실습에 필요한 패키지 설치

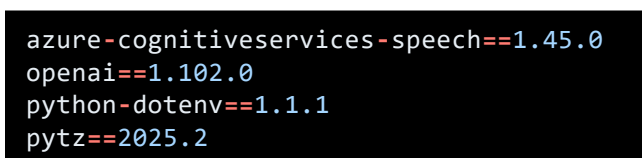
- Explorer 탭에서 New File 아이콘 선택



- requirements.txt 생성



- requirements.txt 에 아래 내용 붙여 넣고 저장



```
azure-cognitiveservices-speech==1.45.0
openai==1.102.0
python-dotenv==1.1.1
pytz==2025.2
```

- requirements.txt 설치
 - Terminal 에서 아래 명령어 실행
 - `pip install -r requirements.txt`

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\soeunpark\OneDrive - Microsoft\soeunpark\07. Study\VSCode>"C:/Users/soeunpark/OneDrive - Microsoft
(venv) C:\Users\soeunpark\OneDrive - Microsoft\soeunpark\07. Study\VSCode>pip install -r requirements.txt

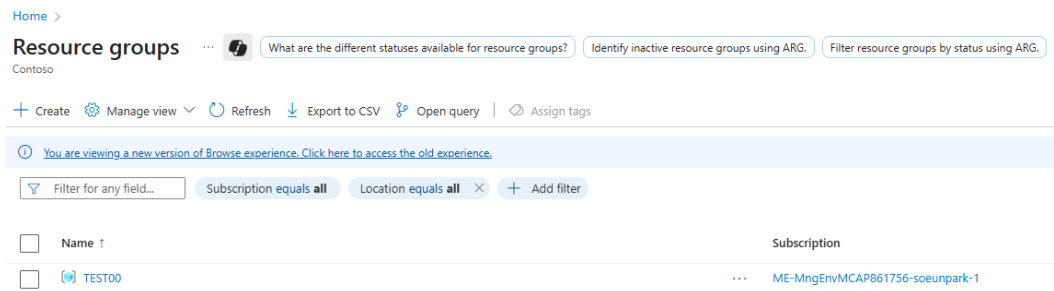
```

기본 환경 설정 및 권한 확인

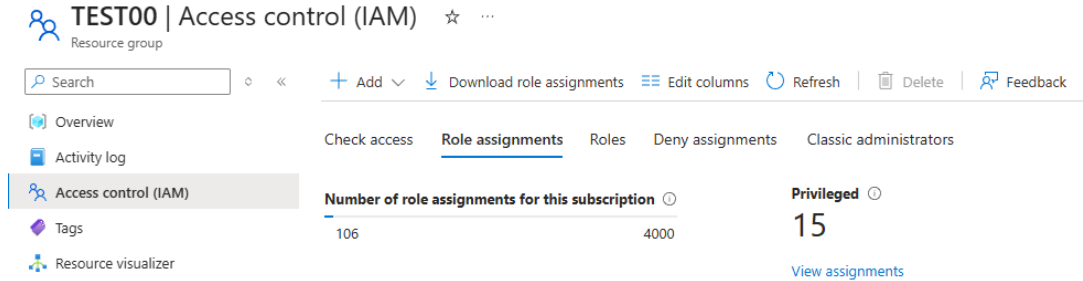
1. Azure portal 접속
 - [Home - Microsoft Azure](https://portal.azure.com) (portal.azure.com)
2. Resource group 생성 확인
 - 아래 아이콘 클릭하여 리소스 그룹 접근



3. 권한 확인
 - 생성된 Resource groups 클릭



- Access control (IAM) > Role assignments 확인



AI Foundry & AI Services 실습

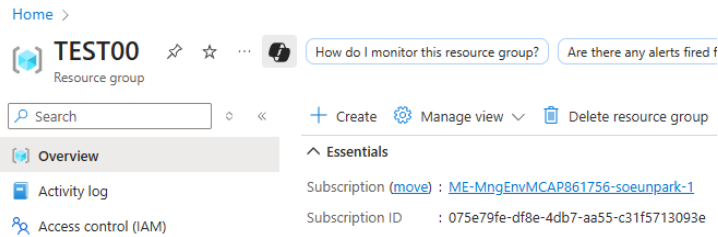
#1 AI Foundry 실습

AI Foundry 프로젝트를 생성하고 간단한 RAG 기반 agent 생성

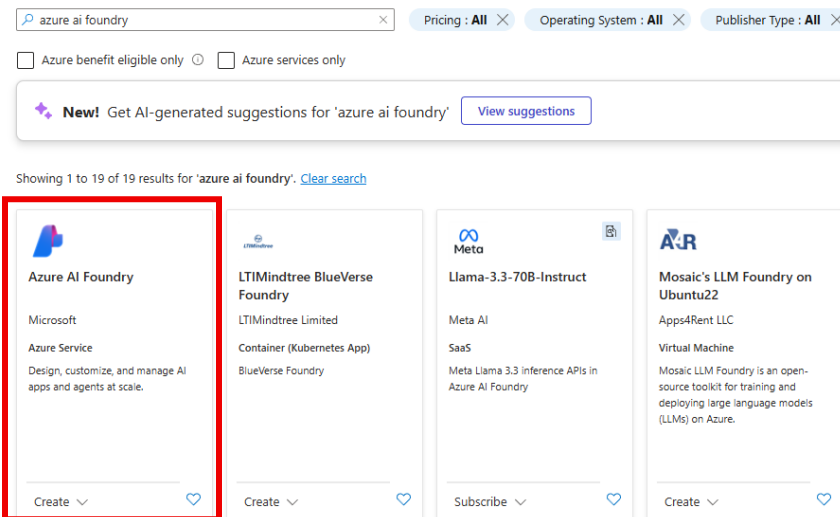
1. AI Foundry 리소스 생성

[Quickstart: Get started with Azure AI Foundry - Azure AI Foundry | Microsoft Learn](#)

- 리소스 그룹 선택 > Create 클릭



- 검색창에 Azure AI Foundry 검색 후 Create 클릭



- 설정 값 입력

[Basics]

Resource group: 생성된 리소스 그룹 선택

Name: <생성할 리소스 이름 입력>

Region: East US

- 나머지 설정은 그대로 두고 Next 클릭
- Review + submit 설정 내용 확인 > Create

Create an Azure AI Foundry resource ...

✓ Basics ✓ Network ✓ Identity ✓ Encryption ✓ Tags **6 Review + submit**

[View automation template](#)

Basics

Subscription: ME-MngEnvMCAP861756-soeunpark-1
Resource group: TEST00
Name: soeunpark-ai-foundry
Region: East US 2
Default project name: firstProject

Network

Inbound Access: All networks, including the internet, can access this resource.

Identity

Identity type: System assigned

- 생성된 리소스 확인 (Azure AI Foundry, Azure AI Foundry project)

☐ Name ↑↓	Type ↑↓	Location ↑↓
☐ SOEUNPARK-FOUNDRY-TEST	Azure AI Foundry	East US
☐ firstProject (SOEUNPARK-FOUNDRY-TEST/firstProject)	Azure AI Foundry project	East US

2. Grounding with Bing Search 리소스 생성

[How to use Grounding with Bing Search in Azure AI Foundry Agent Service - Azure AI Foundry | Microsoft Learn](#)

- 리소스 그룹 선택 > Create 클릭

[Home](#) >

TEST00 Resource group
How do I monitor this resource group? Are there any alerts fired f

Search Create Manage view Delete resource group

Overview

Activity log

Access control (IAM)

Essentials

Subscription (move) : ME-MngEnvMCAP861756-soeunpark-1

Subscription ID : 075e79fe-df8e-4db7-aa55-c31f5713093e

- 검색창에 Grounding with Bing Search 검색 후 Create 클릭

Pricing : All
Operating System : All
Publisher Type : All
Product Type : All

☐ Azure benefit eligible only
 ☐ Azure services only

New! Get AI-generated suggestions for 'grounding with bing search'
 [View suggestions](#)

Showing 1 to 2 of 2 results for 'grounding with bing search'. [Clear search](#)

Grounding with Bing Search

Microsoft

Azure Service

Enhance your AI solution with real-time web search results. To be used as a knowledge tool for Azure AI Agent Service.

Create ▾

Grounding with Bing Custom Search

Microsoft

Azure Service

Create custom domain spaces for AI grounding. To be used as a knowledge tool.

Create ▾

[Basics]

Resource group: 생성된 리소스 그룹 선택

Name: <생성할 리소스 이름 입력>

Region: Global

Pricing tier: Grounding with Bing Search (\$35K per 1K transactions)

- 나머지 설정은 그대로 두고 Next 클릭
- Review + submit 설정 내용 확인 > Create

Create a Grounding with Bing Search resource ...

Basics Tags Review + create

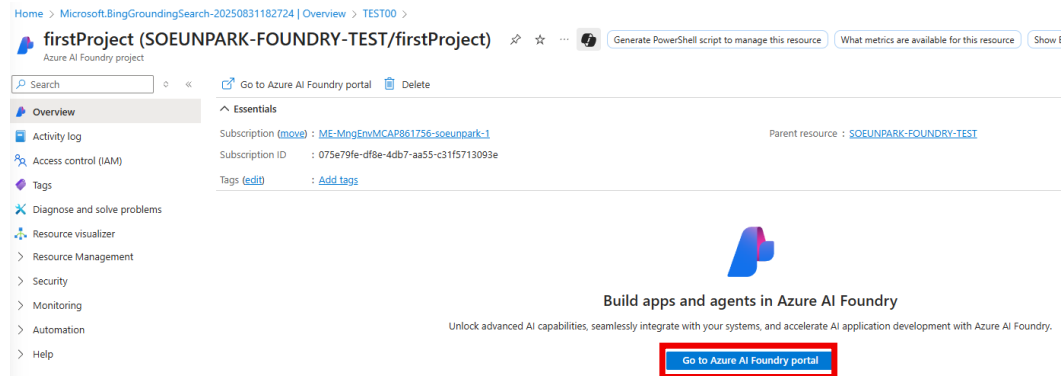
[View automation template](#)

Basics

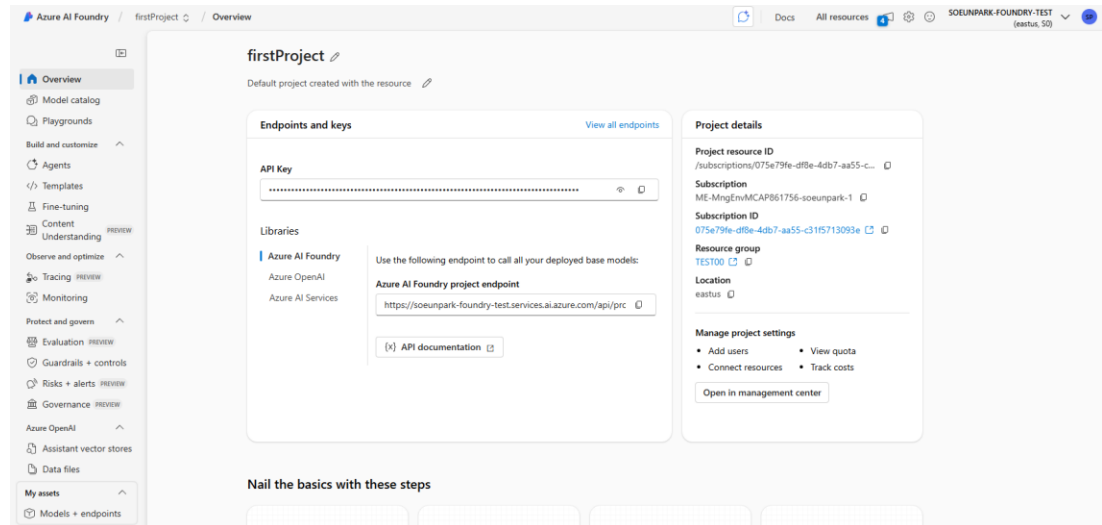
Subscription	ME-MngEnvMCAP861756-soeunpark-1
Resource group	TEST00
Name	SOEUNPARK-BING-SEARCH
Resource group location	East US
Pricing tier	Grounding with Bing Search (\$35 per 1K transactions)

3. AI Foundry portal 접속 (ai.azure.com)

- 생성된 Azure AI Foundry project 리소스 클릭 > Go to Azure AI Foundry portal



- AI Foundry portal 화면 확인

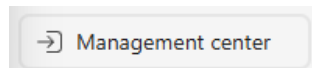


4. Grounding with Bing Search 연결

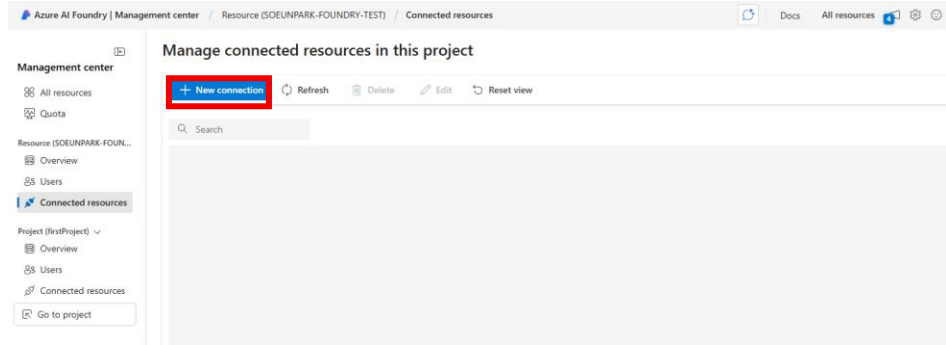
- 메뉴 설명 - All resources / Quota / Overview / Users / Connected resources

- Grounding with Bing Search 연결

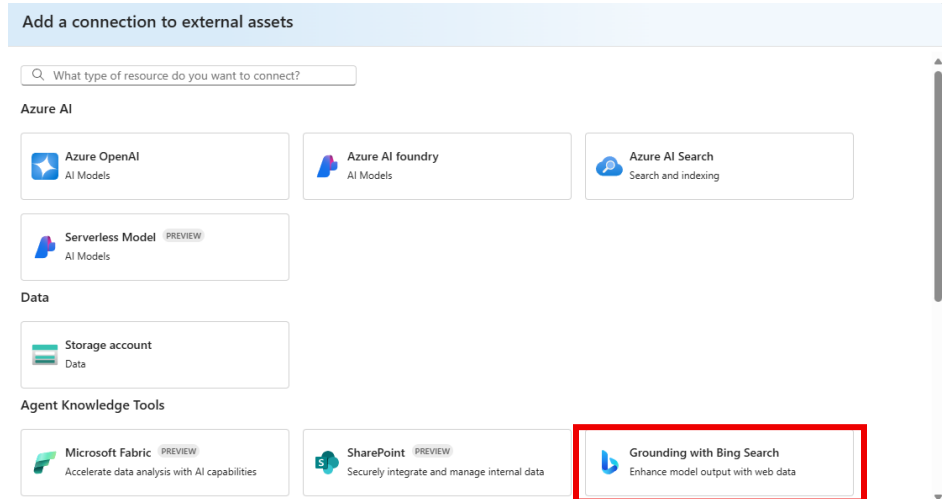
- Management center 클릭



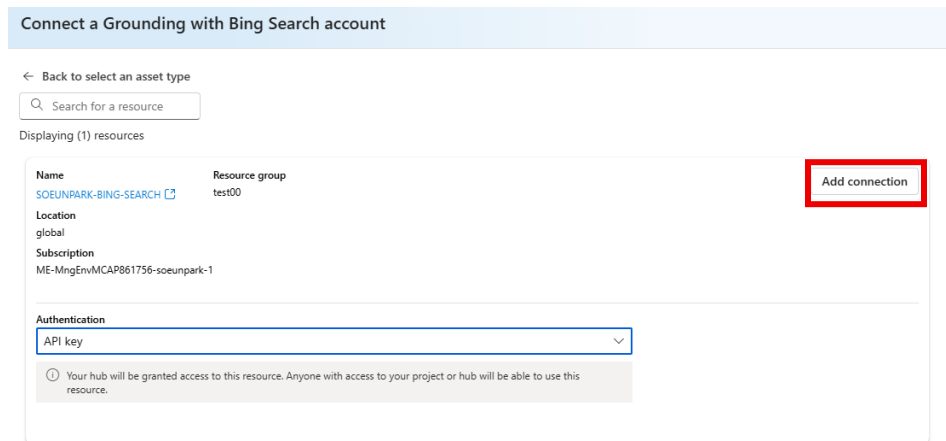
- New Connection



■ Grounding with Bing Search 선택



■ 기존에 생성했던 Grounding with Bing Search 확인 > Add connection



■ 연결 확인

Management center

All resources

Quota

Resource (SOEUNPARK-FOUN...

Overview

Users

Connected resources

Project (firstProject) ▾

Overview

Users

Connected resources

Go to project

Manage connected resources in this project

+ New connection

Refresh

Delete

Edit

Reset view

Search

Name	Access	Authentication type	Added by ▾
SOEUNPARKBINGSEARCH	Shared to all projects ⓘ	API key	soeunpark@microsoft.c...

5. AI Foundry 에 AOAI 모델 배포 – GPT-4.1

- Model Catalog 탐색 – 모델 리스트, Benchmark

Azure AI Foundry / firstProject / Model catalog

Docs All resources SOEUNPARK-FOUNDRY-TEST (testus, 50)

Overview

Model catalog

Playgrounds

Build and customize

Agents

Templates

Content Understanding

Content Understanding

Observe and optimize

Tracing

Monitoring

Protect and govern

Evaluation

Guardrails + controls

Risks + alerts

Governance

Azure OpenAI

Assistant vector stores

Data files

My assets

Models + endpoints

More

Find the right model to build your custom AI solution

Models shown are available for deployment in this project. [See all models](#)

Announcements

Introducing Mistral Document AI (25.05)

Document processing with state-of-the-art OCR and structured data extraction

Check out model Read blog

Introducing GPT-5 models

GPT-5 unifies frontier reasoning and advanced coding with high-performance...

Check out models Read blog

Announcing new gpt-oss models

Push the open model frontier with gpt-oss-120b and gpt-oss-32b, released und...

Check out models Read blog

Model Router with GPT-5

Model router fast-tracks efficient adoption of the GPT-5 models for your business.

Check out model Read blog

FLUX.1 Kontext [pro] | [pro] are now available!

Black Forest Labs founder generation and editing m...

Check out models

Model leaderboards

See what models are performing best in different criteria. [Learn more](#) about our scoring methodology.

Quality ⓘ

1 o3-pro

2 gpt-5

3 o3

Safety ⓘ

1 Phi-4

2 o3

3 o1

Cost ⓘ

1 Ministral-38

2 Phi-4-mini-instruct

3 Phi-4-mini-reasoning

Throughput ⓘ

1 grok-3-mini

2 Llama-3.2-18B-Instruct

3 o3-mini

Collections ▾

Capabilities ▾

Inference tasks ▾

Fine-tuning tasks ▾

Licenses ▾

Search

Models 96

gpt-5-mini

Chat completion

gpt-5-nano

Chat completion

gpt-5-chat

Chat completion

FLUX-1.1-pro

Text to image

FLUX.1-Kontext-pro

Text to image, image to image

codex-mini

Responses

DeepSeek-R1-0528

Chat completion

sora

Video generation

grok-3

Chat completion

grok-3-mini

Chat completion

model-router

Chat completion

o4-mini

Chat completion

MAI-D5-R1

Chat completion

gpt-4.1

Chat completion

gpt-4.1-mini

Chat completion

gpt-4.1-nano

Chat completion

mistral-medium-2505

Chat completion, image classific...

Phi-4-reasoning

Chat completion

Phi-4-mini-reasoning

Chat completion

Llama-4-Scout-17B-16E-L...

Chat completion

- gpt-4.1 클릭 > Use this model

gpt-4.1

[▶ Use this model](#) [Fine-tune](#)

[Details](#) [Benchmarks](#) [Existing deployments](#) [License](#)

The gpt-4.1 series is the latest iteration of the `gpt-4o` model family. This iteration of models is specifically targeted for better coding and instruction following, making it better at handling complex technical and coding problems.

In addition, it increases the context token limit up to 1M input tokens and provides separate billing for small context (128k) and large (up to 1M) context inputs.

As with the previous gpt-4o model family, it supports a 16k output size and features such as:

- Text, image processing
- JSON Mode
- parallel function calling

[See more](#)

- Deployment details 확인 > Deploy

Deploy gpt-4.1

Deployment name *

gpt-4.1

Deployment type

Global Standard

Global Standard: Pay per API call with the highest rate limits. Learn more about [Global deployment types](#).

Data might be processed globally, outside of the resource's Azure geography, but data storage remains in the AI resource's Azure geography. Learn more about [data residency](#).

Deployment details

Model version	2025-04-14	Authentication type	Key
Capacity	140K tokens per minute (TPM)	Resource location	East US 2
Content safety	DefaultV2	Version upgrade policy	Once a new default version is available

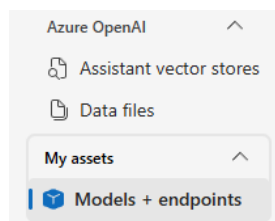
[Customize](#)

Deploy

Cancel

- 배포된 모델 확인

- 왼쪽 메뉴에서 My Assets 밑에 Models + endpoints 선택



- 배포된 모델 리스트 확인

Model deployments Help

Model deployments Service endpoints

+ Deploy model Refresh Edit Open in playground Reset view

Name	Model name	Model version	State	Model retirement date	Content filter	Deployment type
This resource						
gpt-4.1	gpt-4.1	2025-04-14	Succeeded	Apr 11, 2026 9:00 AM	DefaultV2	Global Standard

6. Template 소개 (보여만 주기)

7. Agent 생성

- Agents 메뉴 클릭 > agent 자동 생성

Success: Agent with id asst_bE1ANTv9KNWlv4INxke0Gzst created successfully

Create and debug your agents

My agents My threads

+ New agent Refresh

Search

Name	ID	Model	Created ↓	Description	Tools
Agent411	asst_bE1ANTv9KNWlv4INxke0Gzst	gpt-4.1	Aug 31, 2025 ...		

- Agent 설정

Agent name: <Agent 이름 입력>

Deployment: 이전에 배포한 gpt-4.1 모델 확인

Instructions: You are a helpful assistant. Search for information using the input that I give.

※ Instruction 에 output 형식 지정 등 agent 의 행동 설정 가능

예) Hi, Agent! Draw a graph for a line with a slope of 4 and y-intercept of 9.

Setup Hide

Try in playground

Agent ID ⓘ
asst_bE1ANTv9KNWlv4INxke0Gzst

Agent name
my-agent

Deployment * + Create new deployment
gpt-4.1 (version:2025-04-14)

Instructions ⓘ
You are a helpful assistant. Search information using the input that I give.

○ Knowledge 추가

■ Add 버튼 클릭

Knowledge (0) + Add

Knowledge gives the agent access to data sources for grounding responses. [Learn more](#)

■ Grounding with Bing Search 클릭

Add knowledge

Expand the knowledge an agent has access to by connecting data sources it can use for grounding responses. Currently only a single instance per each type of data source is supported. [Learn more](#)

Add a data source

 **Files**
Upload local files

 **Azure AI Search**
Search and indexing

 **Microsoft Fabric** PREVIEW
Accelerate data analysis with AI capabilities

 **SharePoint** PREVIEW
Securely integrate and manage internal data

 **Grounding with Bing Search**
Enhance model output with web data

 **Grounding with Bing Custom...** PREVIEW
Enhance model output with data from selected web domains

 **Tripadvisor**
Get travel data, guidance, and reviews

 **Morningstar**
Access up-to-date analyst research, expert commentary, and essential Morningstar data

■ 이전에 생성한 Grounding with Bing Search connection 클릭 > Next

Choose an existing Grounding with Bing Search connection

← Back to select knowledge type

ⓘ When you use Grounding with Bing Search, your customer data is transferred outside of the Azure compliance boundary to the Grounding with Bing Search service. Grounding with Bing Search is not subject to the same data processing terms (including location of processing) and does not have the same compliance standards and certifications as the Azure AI Agent Service, as described in the [Grounding with Bing Search TOU](#). It is your responsibility to assess whether use of Grounding with Bing Search in your agent meets your needs and requirements.

+ New connection

↻ Refresh

Search

Filter

Name	Authentication type	Added by	Key
SOEUNPARKBINGSEARCH	API key	soeunpark@microsoft.c...	

[Configure Grounding with Bing Search optional parameters]

Count: 응답으로 리턴할 Search 결과 개수

Set language: 설정 언어(한국어의 경우 ko 입력)

Market: 검색 대상 국가 설정(한국의 경우 ko-KR 입력)

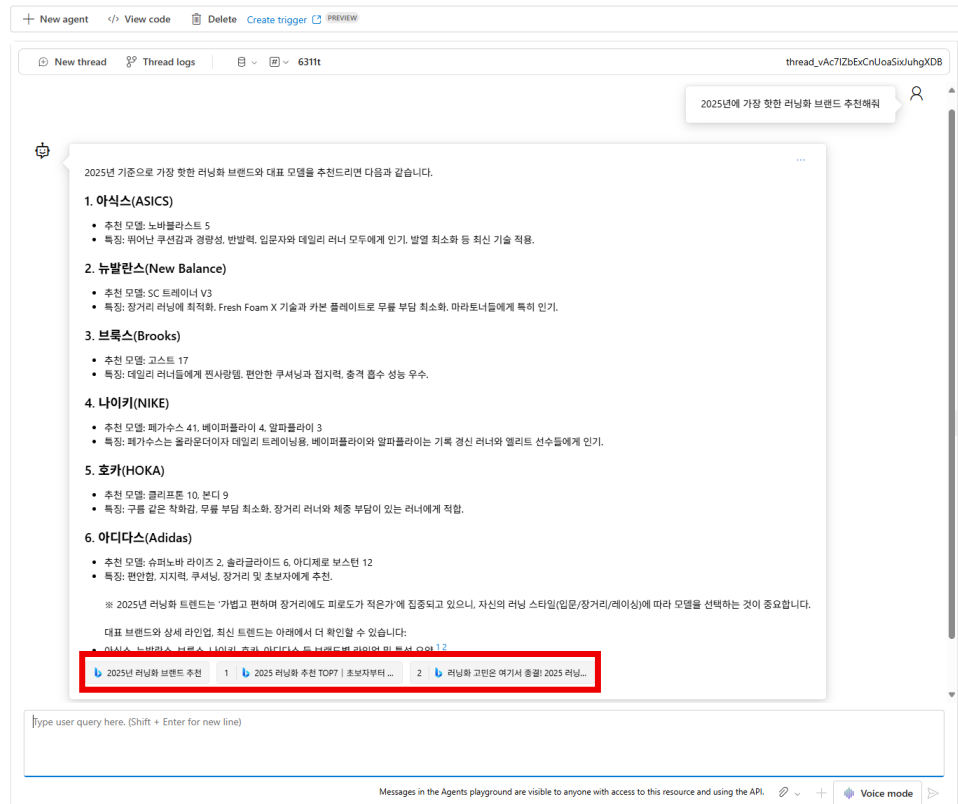
Freshness: 검색 결과 필터링에 사용할 날짜 지정

(2020-09-01~2025-08-31 입력 시 해당 날짜 사이의 결과만 반환)

▪ Try in playground 에서 생성한 agent 테스트

예) 2025 년에 가장 핫한 러닝화 브랜드 추천해줘

← Agents playground



- Voice Agent (보여만 주기)
- Playground 내의 다른 기능 소개 (보여만 주기) – Speech playground > Fast transcription & Speech translation (whatstheweatherlike.wav 파일 사용)

8. Azure AI Content Safety – Prompt Shield / Goundedness / Protected material detection / Custom categories / Text moderation 소개

[Quickstart: Prompt Shields - Azure AI services | Microsoft Learn](#)

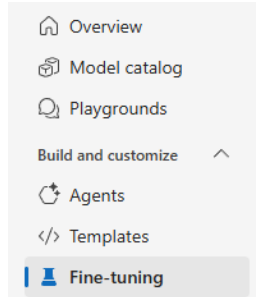
9. Evaluation – Automated evaluation / AI red teaming / Evaluator library 소개

[Evaluate Generative AI Models and Apps with Azure AI Foundry - Azure AI Foundry | Microsoft Learn](#)

10. Fine tuning 실습

[Customize a model with Azure OpenAI in Azure AI Foundry Models - Azure OpenAI | Microsoft Learn](#)

- 왼쪽 메뉴에서 Fine-tuning 선택



- Fine tune model 버튼 클릭

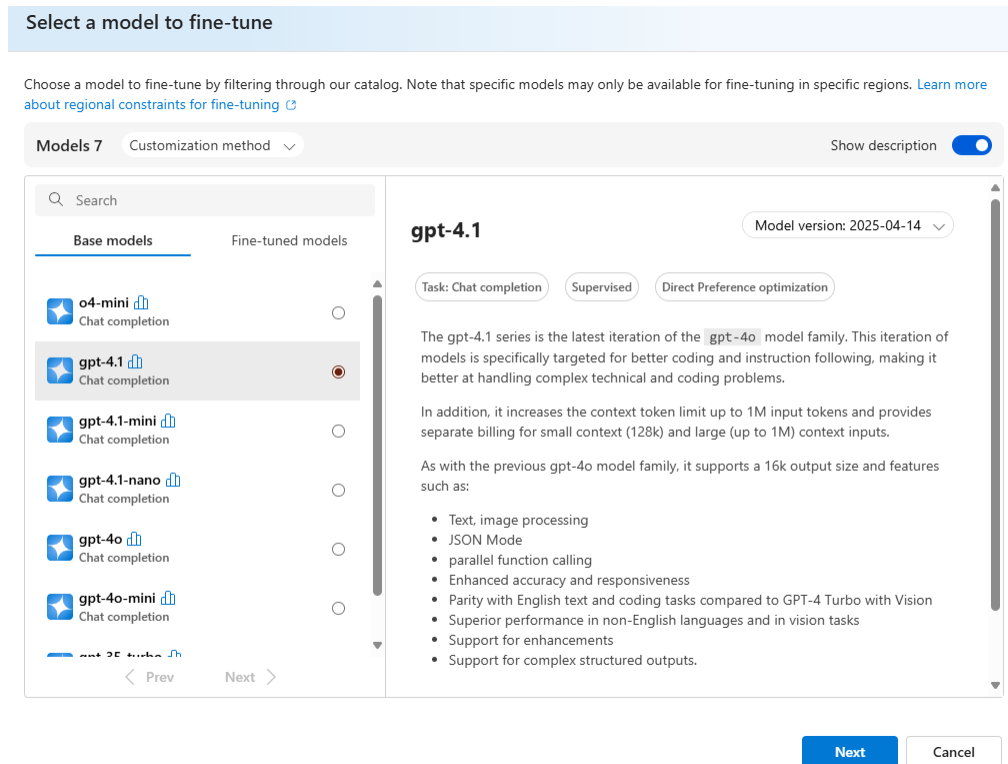
Fine-tune with your own data

Adapt a pre-trained language model to excel at specific tasks or new domains by training it on targeted datasets, teaching the model new skills while retaining

Generative AI fine-tuning AI Service fine-tuning

+ Fine-tune model Refresh Deploy Reset view

- gpt-4.1 선택 > Next



- Training data > Curated sample datasets > textGSM8K_train_sample

Fine-tune gpt-4.1

← Back to fine-tuning

Choose a dataset to customize your model. Ensure your training data is in a .jsonl file, formatted for chat completions, with 'messages' as the row header.

Get started faster - learn how to create datasets using these samples ([chat data](#) , [multi-modal data](#)) on our [GitHub repo](#)

Training data *

Curated sample datasets

★ Curated Sample Datasets

GPT-4.1

Get started with a pre-prepared sample dataset to help you understand how to format data for fine-tuning and test end-to-end workloads. These datasets are intended as examples; they are not intended to produce production quality models.

Select data * ⓘ

textGSM8K_train_sample

☒ Skip preview

[Learn more about preparing your own data](#)

Apply

Cancel

- Completed 확인

Fine-tune with your own data

Adapt a pre-trained language model to excel at specific tasks or new domains by training it on targeted datasets, teaching the model new skills while retaining it's general language

Generative AI fine-tuning

AI Service fine-tuning

+ Fine-tune model

Refresh

Deploy

Continual fine-tuning

Delete

Pause

Cancel

Reset view

Search

Model name	Base model	Status	Customization method	Created on
gpt-4.1-2025-04-14.ft-6cbf87f33cf347b890bb86f	gpt-4.1-2025-04-14	Completed	Supervised	Aug 31, 2025

- 배포된 모델 클릭하여 Details 확인

← gpt-4.1-2025-04-14-ft-6cbf87f33cf347b890bb866decba61c7

Details Metrics Logs Checkpoints

▶ Use this model ↺ Refresh ⚙ Continual fine-tuning 🗑 Delete ⏸ Pause ✕ Cancel

Model attributes

ID
ftjob-6cbf87f33cf347b890bb866decba61c7

Status
✔ Completed

Created on
Aug 31, 2025 9:00 PM

Validation file
-

Method of Customization
Supervised

Auto-deployment enabled
true

Base model
gpt-4.1-2025-04-14

Training file
textGSM8K_train_sample.jsonl

Azure OpenAI resource
SOEUNPARK-FOUNDRY-EU2

Training type
Global (Preview)

Deployment type
Global standard

Training completed on
Aug 31, 2025 10:54 PM

Duration
1h 54m 33s

Training tokens billed
251,000

Deployment Status
[Deployment Successfully Triggered - View Details](#)

Download results Download training file Download validation file

Task parameters

Batch size
1

Learning rate multiplier
2

Number of epochs
3

Seed
367147176

- Fine tuning 된 모델 테스트
 - [Models + endpoints] 탭 접속
 - Open in playground 클릭

Azure AI Foundry / firstProject / Deployments / gpt-4-1-2025-04-14-ft-6cbf87f33cf347b890bb866decba61c7

← gpt-4-1-2025-04-14-ft-6cbf87f33cf347b890bb866decba61c7

Details Metrics Risks & Safety

Open in playground Request quota Edit Delete

■ 테스트 질문 입력

예) 4 명의 학생 중 달리기 선수 2 명을 뽑는 경우의 수는 얼마야?

Chat history

4명의 학생 중 달리기 선수 2명을 뽑는 경우의 수는 얼마야?

Azure AI Foundry | gpt-4.1-2025-04-14.ft-6cbf87f33cf347b890bb866decba61c7

AI-generated content may be incorrect

4명 중 달리기 선수 2명을 뽑는 경우의 수는 조합(Combination)을 사용해 구합니다.

공식: $nCk = n! / (k! \times (n-k)!)$

여기서, $n=4$, $k=2$

따라서 $4C2 = 4! / (2! \times 2!) = (4 \times 3 \times 2 \times 1) / ((2 \times 1) \times (2 \times 1)) = 24 / 4 = 6$

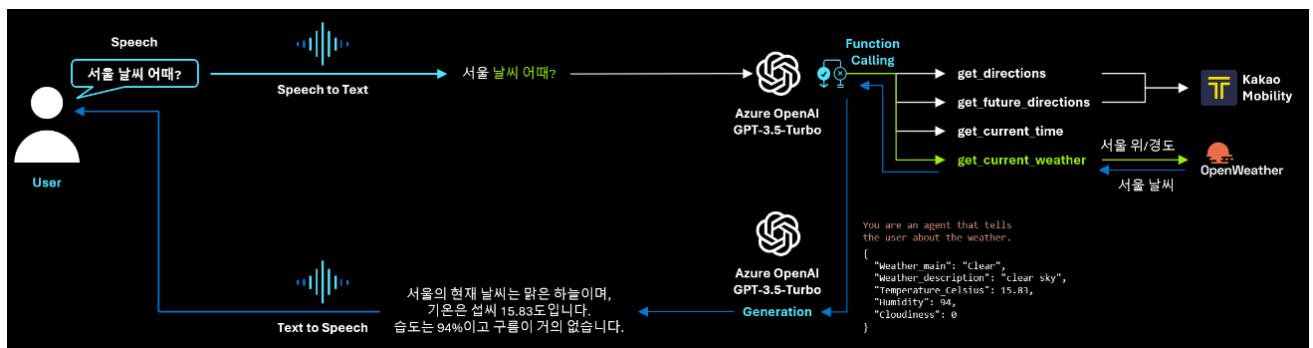
답: 6가지 경우가 있습니다.

#2 나만의 자비스 생성

OpenAI ChatGPT 의 Function Calling 기능을 활용하여, 아이언맨의 자비스와 같은 인공지능 비서 생성

[azure-openai-samples-kr/quick_start/11_my_jarvis_stt_chatgpt_tts.ipynb](https://github.com/azure-openai-samples-kr/quick_start/11_my_jarvis_stt_chatgpt_tts.ipynb) at main · HyounsooKim/azure-openai-samples-kr

- 사용자의 음성은 Azure 가 제공하는 AI Speech 서비스의 STT(Speech-to-text) 기능을 이용하여 텍스트로 변환
- 텍스트는 ChatGPT 가 제공하는 Function Calling 기능을 활용하여, 이미 작성되어진 API 또는 코드를 선택적으로 수행
- 수행 결과는 다시 Azure 가 제공하는 AI Speech 서비스의 TTS(Text-to-speech)를 이용하여 답변

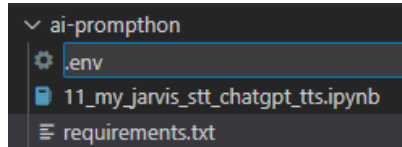


1. 환경 변수 설정 (.env)

- Explorer 탭에서 New File 아이콘 선택



- .env 생성

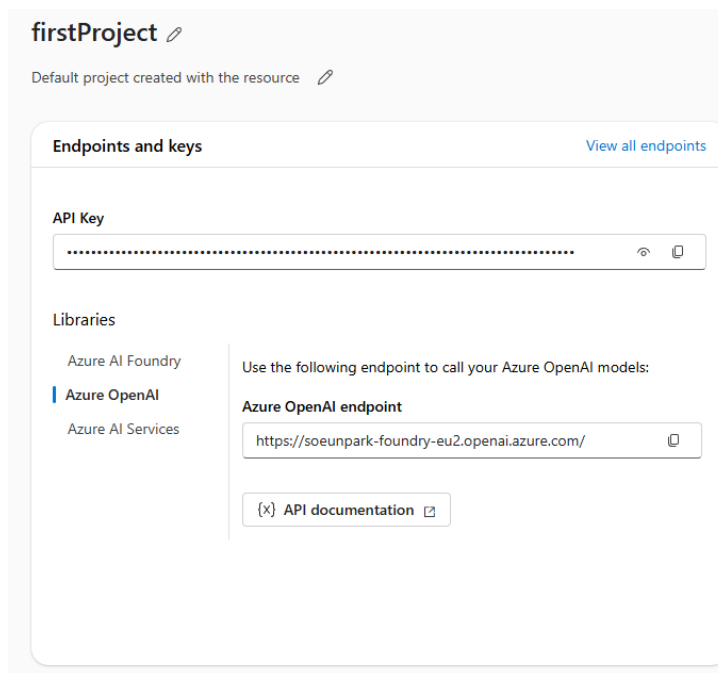


- 아래 내용 .env 에 복사 후 저장

```
AZURE_OPENAI_ENDPOINT=<OPENAI_ENDPOINT>
AZURE_OPENAI_API_KEY=<OPENAI_API_KEY>
OPENAI_API_VERSION=2025-03-01-preview
DEPLOYMENT_NAME=gpt-4.1
AZURE_SPEECH_KEY=<AI_SERVICES_API_KEY>
AZURE_SPEECH_REGION=eastus2
AZURE_SPEECH_LANGUAGE=ko=KR
KAKAO_REST_API_KEY=<KAKAO_REST_API_KEY>
WEATHER_API_KEY=<WEATHER_API_KEY>
```

※AZURE_OPENAI_API_KEY 와 AZURE_SPEECH_KEY 는 같은 값을 사용

- .env 값 가져오기 (Azure AI Foundry portal)



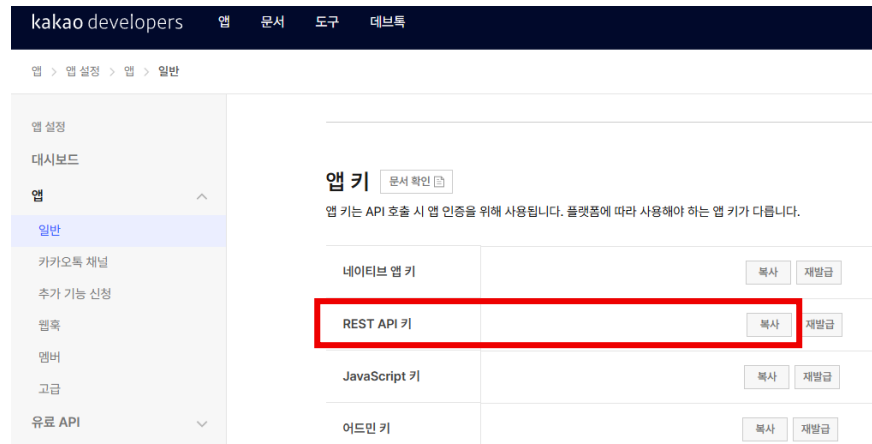
○ KAKAO_REST_API_KEY 발급 방법

[REST API | Kakao Developers 문서](#)

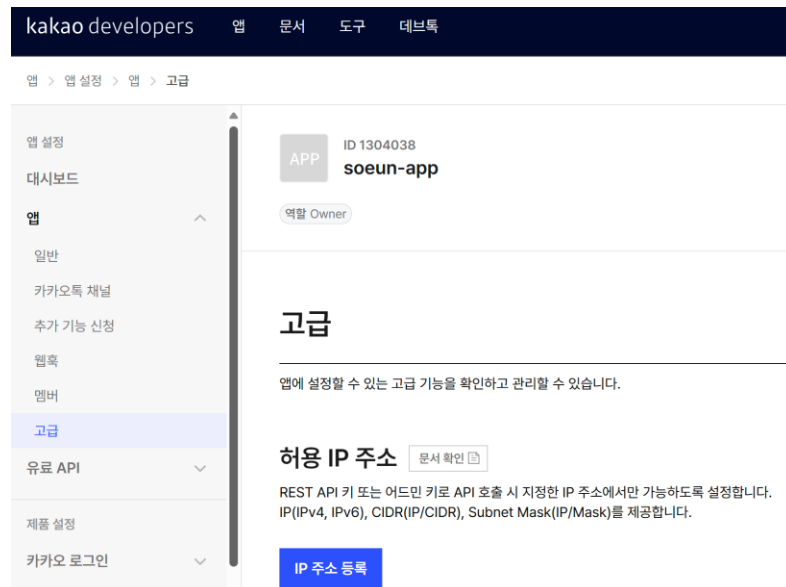
■ 앱 > 앱 생성



■ 생성된 앱 클릭 > 앱 > 일반 > REST API 키 복사



■ 앱 > 고급 > 허용 IP 주소 등록 (REST API 키를 사용하려면 IP 주소 지정 필요)



- 제품 설정 > 카카오맵 > 사용 설정 ON 으로 변경

kakao developers 앱 문서 도구 데브톡

앱 > 제품 설정 > 카카오맵

앱 설정

대시보드

앱

유료 API

제품 설정

카카오 로그인

비즈니스 인증

카카오톡 메시지

카카오맵

푸시 알림

통계

APP ID 1304038

soeun-app

역할 Owner

카카오맵 문서 확인

카카오맵의 주요 설정을 확인하고 관리할 수 있습니다.

사용 설정 문서 확인

카카오맵의 사용 여부를 선택할 수 있습니다.

상태 ☐ OFF

- WEATHER_API_KEY 발급 방법

Current weather data - OpenWeatherMap

- 회원가입 > 메일인증필요
- API > Current Weather Data > Subscribe

OpenWeatherMap Weather in your city

Guide API Dashboards Marketplace Pricing Maps Our Initiatives Blog For Business IMS Support centre

Current & Forecast weather data collection

Current Weather Data

API doc Subscribe

- Access current weather data for any location
- We collect and process weather data from different sources such as global and local weather models, satellites, radars and a vast network of weather stations
- JSON, XML, and HTML formats
- Included in both free and paid subscriptions

Hourly Forecast 4 days

API doc Subscribe

- Hourly forecast is available for 4 days
- Forecast weather data for 96 timestamps
- JSON and XML formats
- Included in the Developer, Professional and Expert subscription plans

Daily Forecast 16 days

API doc Subscribe

- 16 days forecast is available for any location on the globe
- 1-day step for 16 days
- JSON and XML formats
- Included in all paid subscription plans

Climatic Forecast 30 days

API doc Subscribe

- Forecast weather data for 30 days
- JSON format
- Included in the Developer, Professional and Expert subscription plans

Bulk Download

API doc Subscribe

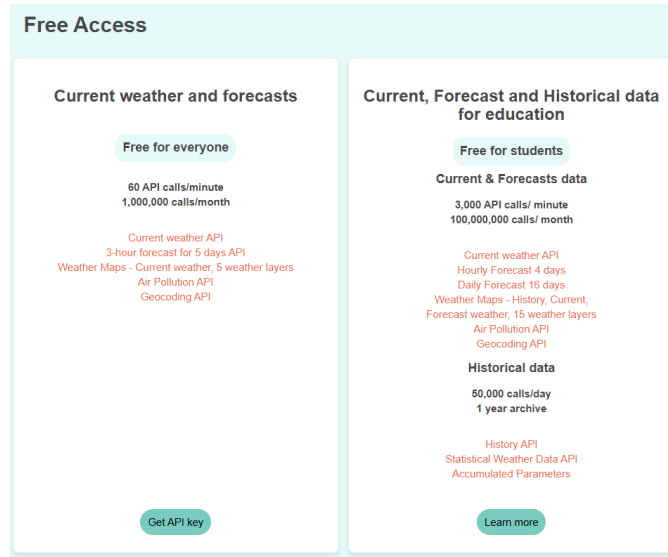
- Current weather, a variety of weather forecasts and their 7-day archive via regularly updated files
- Weather bulks are grouped by types of weather data and location lists (global city lists or ZIP code lists of EU, UK, and US)
- Bulk files are available via CSV and JSON formats
- Included in the Professional and Expert subscription plans

Global Weather Alerts Push notifications

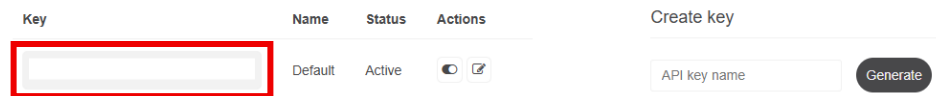
Doc Subscribe

- Get all the **warnings from national weather agencies**
- Weather alerts are pushed to your endpoint
- Data feed provides all active weather alerts from the entire world
- Each alert contains date, time, location, and detailed description
- Monthly subscription**. Please **contact us** to get access.

- 스크롤을 아래로 쪽 내려서 Free Access > Get API key



- Key 값 복사

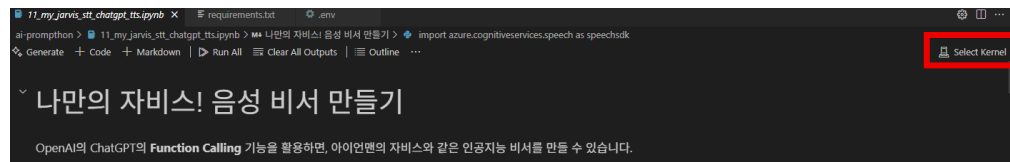


2. Jupyter notebook 코드 다운받기

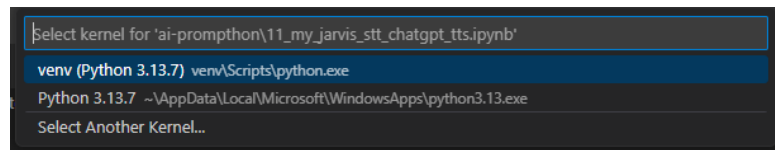
[azure-openai-samples-kr/quick_start/11_my_jarvis_stt_chatgpt_tts.ipynb](https://github.com/HyunsooKim/azure-openai-samples-kr/blob/main/quick_start/11_my_jarvis_stt_chatgpt_tts.ipynb) at main · HyunsooKim/azure-openai-samples-kr

3. VS Code 에서 코드 실행

- 다운로드 받은 Jupyter notebook 선택 > 오른쪽 위에 Select Kernal 클릭



- 이전에 생성한 Python venv 선택



- Shift + Enter 눌러서 코드 하나씩 실행 후 결과 확인

※실행 초반에 ipykernel 설치 팝업이 뜨면 설치 진행